

Institute of Actuaries of India

Subject SA4: Pension & Other Employee Benefits

October 2014 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the paper setters with the aim of helping markers of scripts so as to have a framework and be consistent while evaluating answers. The solutions given are only indicative. It is realized that there could be other points as valid answers and the marker may give credit for any such alternative approach or interpretation which the marker considers to be appropriate.

Solution 1:**i. Reasons for monitoring the financial significance of deviations in experience from the expectation of a defined benefit scheme over a valuation period**

Monitoring of the financial significance of deviations from expectations enables the actuary to:

- Make a semi-independent check on the current valuation results by reconciling them with those from the previous valuation
- Recognise the reasons for any unexpected results and consider the likelihood of future repetition of such results, and advise the client accordingly
- Recognise the potential financial significance of the assumptions chosen for the valuation (and the financial stability of the funding method chosen).
- Reply to questions of Company Management such as HR Head, CFO etc.

It is also a requirement in some jurisdictions that a Scheme Funding report includes an explanation and quantification of the reasons for changes in the scheme's funding level over the inter valuation period.

[3]

ii. Analysis process**Liabilities**

Over a period of time, the Actuarial Liability will change due to the accrual of benefits, the reduction in the period over which the liabilities are discounted, benefit outgo and deviations in experience that affects the liabilities. The Actuarial Liability may also change due to change in Valuation method and/or change in valuation assumptions. The Actuarial Liability (AL) determined using identical assumptions can therefore be expressed by the recurrence relation:

$$AL1 = AL0 * (1 + i) + SCR1 - B1 + LD1$$

Where:

SCR1 = standard Contributions from time 0 to time 1 based on expected experience

B1 = the expected benefit payments between time 0 and time 1

LD1 = the growth in the liabilities due differences between experience and the assumptions.

It will also include the effect due to change in valuation method and/or valuation assumption, if any.

i = valuation rate of interest

Assets

Over the same period, the value of the assets held will be affected by similar factors, except that the deviations will be in experience that affects the investment and contribution income, and the expected contribution payments will be relevant rather than the expected SCR. The accumulating fund of assets can be expressed as:

$$A1 = A0 * (1 + I1) + C1 - B1 + ID1$$

Where:

A1 = assets at time 1

I1 = total expected investment return (investment income + capital gains) from time 0 to 1

C1 = total actual contributions (company + members) paid from time 0 to 1

ID1 = the rise in the asset value due to differences between the expected and actual investment return

Analysing the surplus

The surplus arising during the year can be determined by comparing the surplus at the start and end of the year, i.e.

$$\text{Surplus arising} = (A1 - AL1) - (A0 - AL0)$$

The analysis involves breaking down this surplus in relation to the elements of experience that have deviated from the expectations.

Comments on the Process

The complexity of this process is dependent on the number of elements of experience that are subject to deviation. However, it will usually be possible to remove some of these elements from the analysis on the grounds of financial insignificance.

It is likely, however, that the number of potentially significant elements is sufficient to lead to considerable interaction between the elements and a decision needs to be taken as to how surplus resulting from more than one element of surplus should be allocated in the analysis. There is no single correct way in which the order of items in the analysis should be done.

An example of the interrelationship is the profit (or loss) on investment that are higher (or lower) than expected due to deviations in mortality. This can be counted either in the investment item or in the mortality item of the analysis. The choice on such matters is a matter of personal taste, convenience and practicality as well as the order in which the various factors are investigated.

However, it is usual to start by investigating the effect of assumptions that are expected to have the greatest financial significance. The analysis of some of the minor elements may then be seen to be unnecessary.

The economic assumptions are often those that have the greatest financial significance. However, the demographic assumptions may be very significant if they relate to events that lead to a significant difference in the value of the benefits that are payable, for example if a bulk transfer has been paid or received.

It is important to consider the “overlap” between different factors at all times when carrying out an analysis of surplus. Care should be taken not to double count any aspects or to leave things out accidentally.

Once an item has been analysed, all subsequent calculations are based on the EXPECTED values of that item, to avoid double counting.

The main part of the analysis looks at the inter valuation period and the financial effect of events in that period and hence reconciling the figures at the beginning of the period with

those at the end. If after doing some rough calculations along the lines described above, the results of the valuation are as expected it may not be appropriate to delve much more deeply into the analysis of surplus. The analysis can be a costly and time consuming piece of work and usually the figures cannot be reconciled exactly. The benefit of continuing and refining work on the analysis should be balanced against the costs involved.

[12]

iii. Analysis of surplus

Assumptions

Contributions / benefits / investment returns occur uniformly over the year, approximated to happen all in midyear.

The benefits paid during the year is Rs.60 crores whereas the pensions in payment were Rs.25 crores. In the absence of other benefits, we assume that the excess amount of 35 crores is a lump sum settlement towards commuting the pension – either at the time securing the pension or as a one-time settlement provided for.

The following analysis is approximate to the extent of the information provided in the question.

Workings for Reference – Assets Side

Details	Assets	
Opening Balance	625.00	
Contribution	12.00	
Benefits Paid	-60.00	
Expected Return	48.12	
Investment Surplus	24.88	Item 2
Closing Balance	650.00	

Workings for Reference – Liabilities Side

Details	Actives	Deferreds	Pensioners	Total	
Opening Balance	240.00	100.00	230.00	570.00	
Interest Cost	19.67	8.00	16.05	43.72	
Contribution	12.00	-	-	12.00	
Benefits Paid	-	-	-60.00	-60.00	
Salary Increases	2.56	-	-	2.56	Item 3
Pension in Deferment Increases - Deficit	-	-	-	-	Item 4
Pension in Payment Increases - Deficit	-	-	6.29	6.29	Item 5
Contribution Deficit	3.64	-	-	3.64	Item 6
Unexplained Deficit	2.13	2.00	-2.34	1.79	Item 7
Closing Balance	280.00	110.00	190.00	580.00	
* Service accrual is considered as 15 crores (as the SCR)					
(12 crores as contribution and the balance 3 crores forming part of the contribution deficit)					

1. Surplus brought forward

At the start of the period the surplus was Rs.55 crores. This would have grown to Rs.59.4 [55 * 1.08] crores with an assumed investment return of 8.0% pa.

2. Investment Returns

Expected value of the assets as on 1st April 2014

$$= \text{Rs.}[625 * 1.08 + (12 - 60) * 1.08^{0.5}] \text{ crores} = \text{Rs. } 625.12 \text{ crores}$$

This compares with an asset value of Rs. 650 crores.

Hence, there is a surplus of Rs. 24.88 crores.

3. Salary Increases

The active liability as at 1st April 2014 is Rs.280 crores

Approximate liability allowing for expected salary increases is:

$$= \text{Rs. } [280 * 1.06 / 1.07] = \text{Rs.} 277.38 \text{ crores}$$

Thus the deficit due to salary increases = $\text{Rs.}[280 - 277.38] = \text{Rs.} 2.62 \text{ crores}$

The gain (or surplus) in contributions due to the higher than expected salary increases

$$= \text{Rs.}[12 * (1 - (1.06/1.07)^{0.5}) * 1.08^{0.5}] = \text{Rs.} 0.06 \text{ crores}$$

The net deficit is thus Rs.2.56 crores.

We are assuming that the larger than expected average salary is due to larger than expected salary increases. However, this may not be the case as it could be due to a withdrawals of members with lower than average salaries.

4. Pension in deferment increases

Increases in deferment were as expected, therefore the item of surplus = 0.

5. Pension in payment increases

Impact due to higher increases for pensions in payment

$$= \text{Rs.}[190 * (1.03/1.06 - 1)] = - \text{Rs.} 5.38 \text{ crores}$$

Hence, there is a deficit of Rs. 5.38 crores.

This represents only change in pension liabilities at the current valuation date. We should also consider the liabilities already discharged from the scheme, and the loss we made on those liabilities due to higher than expected pension increases.

For Pensions in payment, this will be

$$= \text{Rs. } [25 * (1.03^{0.5} - 1.06^{0.5}) * (1.08^{0.5})] = \text{Rs.} 0.38 \text{ crores}$$

For Commuted benefits, this will be (assuming commutation considered the higher increases for pensions in payment)

$$= \text{Rs. } [35 * (1.03^{0.5} - 1.06^{0.5}) * (1.08^{0.5})] = \text{Rs.} 0.53 \text{ crores}$$

Thus the total deficit works to $\text{Rs.}[5.38 + 0.38 + 0.53] \text{ crores} = \text{Rs.} 6.29 \text{ crores}$.

6. Contributions

The expected contributions on expected salaries

$$= \text{Rs.}[15\% * 100 * 1.06^{0.5} * 1.08^{0.5}] = \text{Rs.} 16.05 \text{ crores}$$

We have already analysed the salary and investment items, so we must use the expected values for salaries and investment returns in order to avoid double counting this experience.

This compares with the actual contribution income on expected salaries of:

$$\text{Rs.}[12 * (1.06/1.07)^{0.5} * 1.08^{0.5}] = \text{Rs.}12.41 \text{ crores}$$

Thus the deficit in contributions = Rs.3.64 crores

Thus we have:

Details	Surplus	
Opening Balance	55.00	Item 1
Interest Additions	4.40	Item 1
Investment Returns - Surplus	24.88	Item 2
Salary Increases - Deficit	-2.56	Item 3
Pension in Deferment Increases - Deficit	-	Item 4
Pension in Payment Increases - Deficit	-6.29	Item 5
Contribution - Deficit	-3.64	Item 6
Unexplained - Deficit	-1.79	Item 7
Closing Balance	70.00	

7. Unexplained Items

We do not have enough information to analyse the sources of the remaining deficit of 1.79 crores although it appears we are close enough not to further refine the calculations. This could be either due to the demographic experience or changes in the future demographic assumptions. There could be a surplus for example as we have assumed no benefits are paid for actives and deferreds (compared with the expected benefits to be paid).

[14]

iv) Comments

It appears that:

- Investment returns have been better
- The actual investment return works to 12.13% pa higher by about 4.13% pa
- Salary increases have been an item of deficit due to the higher salary increases by 1%
- Higher pension in payment increases have contributed to further deficit
- Pension in deferment increases has not contributed towards surplus or deficit
- Contributions have been an item of loss
- We do not have information above the demographic experience and hence not able to comment on the same

[3]

v) Different Approaches Available

Where there has been a change in the assumptions adopted from those used for the previous valuation, it is possible to perform the analysis using either set of assumptions. However, the assumptions and the values at the current and previous valuation that are used in the analysis must be consistent.

Reworking last time's results on the new basis**(Change of basis item calculated first, at the previous valuation date)**

Under this approach, the structure of the analysis is:

1. Calculate the effect of the change of basis as the difference between the results of the previous valuation on the old basis and the results of the previous valuation reworked using the new basis.
2. Project the reworked results of the previous valuation in line with the new assumptions to calculate the expected position on the new basis at the current valuation.
3. Explain the difference between the expected position on the new basis and the actual position on the new basis.

Calculating this time's results on the old basis**(Change of basis item calculated last, at the current valuation date)**

Under this approach, the analysis can be constructed as follows:

1. Project the results of the previous valuation in line with the previous assumptions to calculate the expected position on the old basis at the current valuation date.
2. Explain the difference between the expected position on the old basis and the actual position on the old basis at the current valuation date – to do this you will need the results of the current valuation on the old basis.
3. Add a final term to explain the change of basis as the difference between the actual position on the old basis and the actual position on the new basis.

Advantages of using the new basis**(Change of basis item calculated first, at the previous valuation date)**

Reasons in favour of performing the analysis using the assumptions that have been adopted for the latest valuation include:

- It provides a better check on the new valuation results
- To the extent that any changes in the assumptions reflect experience over the inter valuation period, the actual and expected results will be closer together and there should therefore be fewer financially significant factors and a reduction in the amount of work and accuracy required
- The results of the analysis may be more useful in indicating the financial significance of deviations from the current assumptions and so help in projections of the effects of future deviations.

Advantages of using the old basis**(Change of basis item calculated last, at the current valuation date)**

However, in many cases it may not be practical to rework the previous valuation on the new assumptions (for instance if the previous valuation was carried out by a different actuary) and so it may be necessary to perform the analysis using the assumptions adopted at the previous valuation and to rework the results of the current valuation on that old basis. The change of basis would then be a final alteration.

This approach has three main advantages;

- It may help to set the new basis since financially significant departures from the old basis will be revealed
- It may help the trustees understand the significance of the change in assumptions

- It may not be possible to rework the old valuation on the new basis, e.g. if computer files have been lost or corrupted or if this is the first time that your organisation is performing a valuation of this scheme.

[8]

[40 Marks]

Solution 2:**(i) Retirement Trust:**

A trust is an equitable obligation binding people (called trustees) to deal with property (called trust property) over which they have control for the benefit of people (called beneficiaries), any one of whom may enforce the obligation. There must be trust property for a trust to exist.

Parties involved in a pension scheme trust are:

- ☐ The trustees
- ☐ The members and beneficiaries
- ☐ The sponsoring (or principal) employer
- ☐ Any participating employers whose employees can also join the scheme.

[2]

(ii) EMPLOYEES PROVIDENT FUND:

The Employees Provident Funds and Miscellaneous Provisions Act, 1952 was enacted to provide a kind of social security to the Industrial Workers. It mainly provides retirement or old age benefits such as Provident Fund, Superannuation Pension, Family Pension and Deposit Linked Insurance.

Employers employing 20 or more employees have to contribute towards Employees' Provident Fund (EPF) Scheme. The Scheme is administered by Employees' Provident Fund Organisation (EPFO) through Board of Trustees.

It is a defined contribution benefit where contributions are made by both the employer and the employees in equal proportion with voluntary additional contribution by the employees during the working lifetime of employees. The accumulated value as a lump sum is available to the employee (or beneficiary, as the case may be) to provide him the economic security in retirement (or death of the employee). The Rules and Regulations of the Scheme are given in the Employees' Provident Fund Scheme, 1952.

Contributions: Employer contributes every month at the rate of 12% of wages and an equal amount is contributed by the employees. The wages include basic wages, dearness allowance, cash value of any food concessions and retaining allowance. An employee can make additional contributions upto 12% of wages optionally.

Benefits (Interest and payments):

Interest is credited to the member's account at a rate declared by the Central Government in consultation with the Central Board of Trustees. Till 31 March 1993 interest was credited on the amount standing to the credit of a member as on 1st day of April every year. From 1 April 1993 the yearly interest is computed on monthly running balances.

Full accumulated amount may be withdrawn by the member on normal retirement from service (after attaining the age of 55 years) or on retirement on account of permanent & total disablement, voluntary retirement etc. In case of death of the member, the amount will be paid to the nominee. Ninety percent of the amount can be withdrawn by the member within one year before the retirement or after attainment of age of 59 years, whichever is earlier.

Non-refundable withdrawals are allowed for the following reasons such as housing and insurance (i.e. financing life insurance policy on member's own life), illness in certain cases (involving hospitalisation etc.), marriage or post-matriculation education of children etc. However, such withdrawals are made subject to certain conditions and within certain limits.

Funding vehicles: The Company can choose to pay the PF contributions to the Regional Provident Fund Office or have an exempt arrangement. Appropriate Government may exempt an establishment under Section 17 of the Act if the rate of contribution is not less than that of Employees' Provident Fund Scheme and at the same time, the benefits on the whole are not less favorable than those of EPF Scheme. The terms and conditions of exemption are given in Appendix A of the Act.

Income Tax Provisions:

Provident Fund is a nearly tax-free build-up retirement provision as is clear from the following:

- Employer's contribution under Section 36(1) (iv) of the IT Act is treated as a deduction in the computation of profits and gains of the business or profession of the employer.
- The employer's contribution to PF is not treated as income of the concerned employee, if it does not exceed ten or twelve percent of his salary.
- An employee's own contributions shall be deducted under Section 80C (iv), (v) and (vi) of IT Act up to one lac fifty thousand along with other investments such as life insurance, annuities, PPF etc. Income of the Trust (interest and capital gains) is exempt from tax as per section 10(25) (ii) of IT Act.
- Interest allowed to individual member's accounts at a rate in excess of that notified by the Government will become taxable in the hands of the employees as per Rule 6 of Part A of the Fourth Schedule of IT Act.
- In terms of Section 10(12) of Income-Tax Act, the accumulated balance payable to the employee is tax-free to the extent provided in rule 8 of Part A of the Fourth Schedule. However, if the balance is withdrawn after putting in less than five years continuous service, then the same may be taxed except where the termination of service is due to ill-health or by contraction or discontinuance of employer's business

or other cause beyond the control of the employee. In case of transfer to other recognised Provident Fund also, no tax is payable.

- Hence at present provident fund is among some of the savings instruments which are exempted from tax at all the three stages, i.e. contribution, accumulation and benefit delivery.

Investment pattern for Employees' Provident Funds:

S. No.	Investment Pattern	% amount to be invested
(i)	(a) Government securities ^I	Up to 55
	(b) Other securities ^{II} , the principal whereof and interest whereon is fully and unconditionally guaranteed by the Central Government or any State Government except those covered under (ii) (a) below.	
	(c) Units of mutual funds set up as dedicated funds for investment in Government securities and regulated by the Securities and Exchange Board of India; Provided that the exposure to a mutual fund shall not be more than 5% of the total portfolio at any point of time.	
(ii)	(a) Debt securities with maturity of not less than three years tenure issued by Bodies Corporate including banks and public financial institutions ^{III}	Up to 55
	Provided that at least 75% of the investment in this category is made in instruments, having an investment grade rating, from at least one credit agency.	
	(b) Term Deposit Receipts of not less than one year duration issued by scheduled commercial banks subject to certain conditions.	
	(c) Rupee Bonds having an outstanding maturity of at least 3 years issued by Institutions of the International Bank for Reconstruction and Development, International Finance Corporation and the Asian Development Bank.	
(iii)	Money market instruments including units of money market mutual funds	Up to 5
(iv)	Shares of companies on which derivatives are available in Bombay Stock Exchange or National Stock Exchange or equity linked scheme of mutual funds regulated by the Securities and Exchange Board of India.	
	Nil	

GRATUITY:

The Central Government enacted the Payment of Gratuity Act, 1972 (Gratuity Act) which has come into force with effect from 16 September 1972. It was with the object of providing a uniform scheme of payment of gratuity to industrial workers throughout the country.

Gratuity Act applies to nearly all employers who employ (or had employed on any day preceding 12 months) ten or more employees. It becomes obligatory on the part of such employers to pay gratuity to their employees. Now the Gratuity payable as per the Gratuity Act (Act Gratuity) is a statutory benefit for all employees irrespective of their salary.

Benefits:

As per Gratuity Act, the gratuity is computed at the rate of 15 days' wages for each year of service. For this purpose a month's wages are treated as 26 days' wages and hence the benefit accrues at the rate of 15/26 of monthly wages for each year of service. The Act gratuity is thus calculated as under:

Act Gratuity = $15 / 26 * \text{Service in years} * \text{monthly wages}$

A part of service in excess of six months is taken as one year and wages are the last drawn monthly wages of the employee concerned. Further, in case of seasonal workers, the gratuity is calculated at the rate of 7 days' wages instead of 15 days' wages.

Gratuity is payable immediately on the "termination" of employment of the employee after he has rendered a continuous service of not less than five years:

- a. on his superannuation; or
- b. on his retirement; or resignation; or
- c. On his death; or disablement due to accident or disease.

The completion of continuous service of five years is not necessary in case of death/disablement.

As per the Act, the amount of gratuity shall not exceed ` Ten lacs. This ceiling has been revised from time to time. Present ceiling on gratuity amount is applicable with effect from 24 May 2010. Earlier to it the ceiling was ` 3.50 lacs. The Gratuity Act however does not deny the right of an employee to receive better terms of gratuity under any award or agreement or contract with the employer. An employer may therefore pay gratuity at a rate of more than fifteen days' wages (e.g. one month wages) for each year of service with or without any ceiling. The Act merely prescribes a minimum gratuity and therefore no employer can provide less than the Act benefit.

Taxation Aspects:

The gratuity as and when paid by the employer to his employee, is deductible from his income under section 37(1) of Income Tax Act, 1961. It is because in terms of Section 17(1) (iii) of the Act, gratuity is a variant of salary and hence for the employer it is a necessary business expense. For the recipient (i.e. employee) it is taxable under Section 15 of the Act.

However, as per Section 10(10) (ii) of IT Act any gratuity received under the Gratuity Act is not taxable. Also as per Section 10(10) (iii) of IT Act any gratuity which does not exceed one half month salary for each year of completed service, subject to such limit as notified by the Central Government is not taxable.

The Government has encouraged the employers to fund their gratuity liability by providing a number of tax-incentives. Such tax benefits are available to Gratuity Funds provided they are approved by the Commissioner of Income Tax under Part C of Fourth Schedule of Income Tax Act, 1961 and Part XIV of Income Tax Rules, 1962. The conditions to be satisfied by a scheme to qualify for approval are set out in Rule 3 of the said Schedule. The main conditions include that the Scheme should be run by the Trustees, the trust should be irrevocable in nature, the employer should be contributor to the fund, funding of gratuity should be the sole purpose of the fund, contributions should not exceed 8.33% of salary etc.

An approved Gratuity Fund is entitled to following Tax reliefs:

- a. As per Section 36(1) (V) of IT Act contributions paid by the employer are treated as business expenses of management.
- b. As per Central Board of Direct Taxes (CBDT) circular No. 14 dated 23.4.1969, if the past service liability is funded by the employer in a single installment or in maximum five installments, the whole of the contribution made by the employer will be treated as business expense in the year of payment.
- c. The contributions (or premium in case of an insured Scheme) will not be treated as income in the hands of the employees and hence will not be taxed in their hands. This is as per a directive by the CBDT in which a ruling was given that the premia or the amounts payable by the Trustees of an approved Gratuity Fund to a life insurer or life insurers for obtaining a Group Gratuity Scheme will not be regarded as taxable perquisites under Section 17(2) (V) of Income-tax Act in the hands of the employees.
- d. Income (whether interest or capital gains) out of the investments of the Trust is not taxable as per Section 10(25) (iv) of IT Act.

Funding vehicles and investment pattern:

Approved Gratuity Trust Funds are irrevocable in nature where the employer does not have the proprietary control over the funds. The gratuity rights of the employees become independent of the fortunes of the business as they are invested outside the business. These Schemes are managed by the Trustees who look after the interest of the members and have control over the funds.

The Trustees may either invest the funds themselves (in which case the scheme is called privately (or self) managed Scheme) or may adopt a gratuity scheme of a life insurance company which could be traditional funds or unit linked schemes.

In case of a privately managed Scheme the funds are to be invested as per the pattern prescribed under Rule 101 of the IT Rules. The Trustees have to set up machinery to deal with day-to-day administration such as investment of funds, collection of interest and maturity proceeds, maintenance of records and books of accounts, sale of securities, and payment of benefits to members etc. They have also to arrange for periodical actuarial valuation and annual audit of accounts of the Trust. The scheme, whether it is self-managed or insurer administered, has to be got approved by the Commissioner of Income tax in order to avail tax benefits.

Investment pattern for Gratuity (and Superannuation) schemes is as under:

- i.** Government securities & Government guaranteed securities
upto 55%
Including units of mutual funds set up as dedicated funds for
investment in Govt. securities
- ii.** Debt securities with maturity of not less than 3 years issued
upto 40%
By body corporate including banks & public financial institutions
- iii.** Money market instruments
upto 5%
- iv.** Shares of companies
upto 15%

GROUP SUPERANNUATION:

Superannuation is a non-mandatory corporate pension scheme offered to employees. Although it is not mandatory to provide pension to their employees, the tax incentives offered by the Government have played a great role in the steady growth of superannuation schemes in India. One of the objectives for introduction of superannuation benefits is also to enable long term savings and income adequacy at retirement. It may either be a defined benefit scheme or a defined contribution scheme.

Set up, Taxation Aspects, contribution and benefits:

The Government offered attractive tax incentives for providing superannuation benefits. However, such incentives are available if benefits are funded through a superannuation scheme approved by the Commissioner of Income-tax under Part B of Fourth Schedule of Income-Tax Act and Part XIII of Income-tax Rules. The main conditions to be satisfied to qualify for approval are that the scheme should be run by the trustees and the trust should be irrevocable in nature, i.e. no moneys belonging to the trust shall be receivable by the employer under any circumstances.

The employer should be a contributor to the fund and its sole purpose should be to provide for annuities to the employees on their retirement on or after a specified age or on their becoming incapacitated prior to such retirement or for their widows, children or dependants on death.

There is also restriction for a Director of the company to be admitted to the benefits of the fund. Only those directors may get benefit under the scheme who are whole time bonafide employees of the company and do not own more than five percent voting power shares of the company.

Contribution made by an employer up to ` one lac is not treated as income in the hands of the employee. Any contribution over and above this sum for an employee is taxable income for that employee Under Section 17 (2) (vii) of I T Act.

Section 36(1) (iv) of Income-tax Act allows the ordinary annual contribution paid by an employer as a deductible expense from his income for the purpose of computation of profits and gains of the business.

As per rule 89(ii) of Income-tax Rules, it is obligatory on the part of trustees to purchase annuities from life Insurers. However, till the pension comes into payment the trustees may accumulate contributions in which case they have to be invested as per Rule 85. The investment pattern is as stated above Under Gratuity.

Income of the Fund (both interest and capital gains) is exempt from Income-tax under Section 10(25) (iii) of Income-Tax Act.

Pension is treated as “Salary” income under Section 17(1) (ii) of Income-tax Act and thus annuities payable to employees from Approved Superannuation Funds are taxable in their hands as salary. However, commuted value received on retirement at or after a specified age or on his becoming incapacitated prior to such retirement is tax free under Section 10(13)(ii). As per Rule 90 of Income-tax Rules the maximum commutation allowed is one-third of annuity, if the employee is entitled to receive gratuity, or one half, in other cases.

NATIONAL PENSION SYSTEM:

The government set up an interim regulatory authority viz. Pension Fund Regulatory and Development Authority (PFRDA), on 23rd August 2003 to regulate and supervise pension funds and also to develop pensions in India. The Government also introduced PFRDA Bill, 2005 which has resulted into an Act in 2014 mandating PFRDA to act as a Regulator and develop the Pension Sector in India. PFRDA has set up a Trust under the Indian Trusts Act, 1882 to oversee the functions of the Pension Fund Managers (PFMs).

The New Pension Scheme (NPS) will be voluntary in general but is compulsory for Central Government employees who join service on or after 1 January 2004.

Government has announced that NPS would be available to every citizen from 1 April 2009 on a voluntary basis. It is a defined contribution scheme open to any citizen of India, whether resident or non-resident aged above 18 and below 60 years. The Contributions are flexible and may vary from month to month or year to year. There is no upper limit for contribution amount and the number of contributions in a year. However, tax exemption limits apply and minimum contribution in a year is INR 6,000.

Further a scheme known as Swavalamban Scheme has been introduced for the unorganized sector of India for those who are not Government or Semi-Government employees and are not covered under EPF Act. The Government will be contributing ` 1,000/- per annum for three years to their account provided such people contribute at least ` 1000/- and at the maximum ` 12,000/- per annum. This has been done in order to encourage people in unorganized sector to contribute for their old age.

A NPS Corporate Model was launched late 2011. It is a voluntary defined contribution pension scheme that can be provided by employers. Under a corporate NPS scheme set up by an employer, employer contributions up to 10% of Basic Salary would be a tax deductible expense. Also contributions up to 10% of Basic salary, paid by the employer on behalf of the employee would not be treated as a perquisite.

An employee's own contributions would be eligible for tax exemption in line with other long term investment allowance up to Rs 150,000.

Under a corporate NPS scheme, the company applies for a corporate NPS ID and the scheme can be optional or mandatory for all employees. Companies liaise with a Point of Presence (POP) to register employees and employees would have individual NPS accounts. Each account will have a company ID number attached in addition to the individual account number. Continuation of the account is immediate after leaving the employer.

The investment model allows choice for either the employer to choose the allocation or leave it to employees to choose. Also there are two investment options available called active choice and auto choice. Under active choice, the subscriber can specify the percentage in which his / her money is to be invested in the three asset classes viz. Government securities (allowed up to 100%), corporate bonds (allowed up to 100%) and equities (allowed up to 50%). Under auto choice, the system will automatically calculate the asset allocation percentages based on the subscriber's age i.e. a graded % based on age, starting from age 35. For example, at age 55, 80% would be invested in government securities and 10% each in the other two asset classes.

A portion of the accumulated balance at retirement is available as a lump sum and the balance has to be annuitized mandatorily. Currently for withdrawals before age 60, up to 20% can be drawn as a lump sum and the balance 80% should be annuitized. Also at age 60, up to 40% can be drawn as a lump sum. However if the accumulated lump sum is less than INR 200,000, then the same can be drawn as a lump sum with no annuitisation restrictions. Non-refundable withdrawals of up to 25% of the accumulated amounts have been allowed for health reasons.

In addition the lump sum at age 60 can be deferred by up to three years and likewise the annuity draw down can be deferred until age 70 to protect the members against volatility in the investment markets.

While the withdrawal restrictions apply to Tier I accounts, there is an additional account that has no such restrictions called Tier II.

Interest earned on investments is tax free in the hands of the trust/employee. However the proceeds from NPS either in the form of lump sum or annuity post retirement is taxable. Return would depend on performance of underlying portfolio, choice of investment pattern and fund manager available.

The following stakeholders are a part of the NPS structure.

1. Trustee bank
2. Central Record keeping Agency (CRA)
3. Point of Presence (POP)
4. Pension Fund Manager (PFM)
5. Custodian &
6. Annuity service provider

[30]

[32 Marks]

Solution 3:

(i) Risks to the Sponsor:

A defined benefit scheme is one in which benefits are defined in advance but costs are uncertain. Costs depend on the level of benefits and on future experience. Contributions are estimates of the amounts to be set aside to meet the cost of future benefits and hence cost and contributions are different.

The three main risks for a Defined Benefit scheme are:

1. Sponsor covenant
2. Investment risk
3. Longevity risk

Sponsor covenant is the combination of the ability and willingness of the sponsor to pay (or the trustees to require the sponsor to pay) sufficient contributions to meet the benefits as they fall due.

Investment risk can be interpreted in a number of ways. For example, mismatching, default, volatility, marketability, liquidity, diversification, high expenses and changes in taxation could all be important investment risks. Inflation (including salary inflation) can also be a risk as the liability may/will depend on it.

The risk of choosing assets that are not appropriate for meeting the liabilities is likely to be a particularly important risk for defined benefit schemes. In order to choose a suitable investment strategy, it may be appropriate to build a model of the asset and liability cash flows i.e. an asset-liability model.

Another important risk for defined benefit schemes is funding risk, the risk that insufficient funds are built up to meet the liabilities as they fall due.

If the scheme pays annuities directly from the trust, then improving longevity would mean increased costs to the sponsor. If in case, annuity is purchased, the purchase terms may deteriorate due to some or all of the following factors:

1. A reduction in interest rates.
2. Expectations of longevity increasing.
3. Competition in the annuity market reducing.
4. Insurers' capital requirements for annuity policies increasing.

Additional risks include operational, legal and political risks. Operational risks include:

1. inappropriate advice given to employer or trustees, resulting for example from:
 - incompetence or insufficient experience of the advisor,
 - lack of integrity of the advisor, perhaps due to sales-related payments,
 - the use of an unsuitable model or parameters,
 - errors in the data relating to the beneficiaries
2. Poor record keeping leading to inefficiencies and inaccurate calculations.
3. high running costs, e.g. professional fees and administration costs, perhaps as a result of changes in legislative requirements
4. incorrect benefit payments/ over payment - for example, continuation of pension payments after a pensioner has died
5. Late investment of contributions, perhaps resulting in loss of investment return
6. Fines or penalties for non-compliance with legislation
7. Expense risks

Risks of legislative changes, for example:

- Benefit improvements required by law that could increase costs.
- Greater certification and more procedures to follow, eg disclosure of information to members, resulting in increased running costs.

Also the sponsor would face the risk of insufficient protection for early deaths and disability and hence the need for provision of adequate covers for such cases.

[8]

(ii) Options for mitigating such risks for the sponsor:

The following actions could be taken to manage risks associated with sponsor covenant and insufficient funds:

- Consider introducing more prudence in the funding basis, leading to surplus building up with favorable experience.
- Conversely, can some elements of prudence be reduced, without compromising the desired level of security (particularly by the trustees) to avoid excessive funding requests that may lead to sponsor insolvency and

- More frequent monitoring of the plan's funding and / or solvency position may be appropriate.

To mitigate investment risks:

1. The investment manager's performance can be reviewed to make sure good performing managers are used.
2. An asset–liability study can be undertaken to review investment strategy in relation to the plan's liabilities.
3. Pensions in payment may be bought out to insure against adverse investment and mortality risk, although this may be prohibited by the associated costs.

To manage operational, legislative and political risks:

1. A complete independent audit of the administration function to track procedures and identifying potential pitfalls and areas for reform.
 - Draw up a complete procedures manual, eg to cover: receipt of contributions and their investment, benefit and options calculations.
2. Strive for best practice and not just the legislative minimum.
3. Ensure that checking and authorisation procedures exist and are followed.
4. There should be regular reporting by the administrator, fund manager and other key advisers to the trustees to ensure that the plan is being operated correctly.
5. Ensure sample checks of any calculations are carried out by the actuary.
6. Review implementation of recent legislative changes and requirements.
7. Check that all the plan's documents are up to date, eg trust deed and rules, employee booklet.
8. Verify consistency of the trust deed and rules, booklets and announcements.
9. Verify that recent decisions taken comply with the trust deed and rules.
10. Check that all those involved with the day to day running of the plan have been adequately trained. If not, arrange training.
11. Arrange for trustee insurance, if possible, against for example any claims for breach of trust.

Also

1. Review cost neutral terms for individual benefit options and limit the risk of financial selection against the plan by requiring company consent.
2. Review insurance arrangements, in particular insurance of death-in-service benefits in the event of a catastrophe

[8]

(iii) Employees' Pension Scheme (EPS):

Employees' Pension Scheme, 1995 (EPS) has been introduced with effect from 16.11.1995. Prior to this, Employees' Family Pension Scheme, 1971 was in operation since April, 1971. The Scheme applies to all those establishments to whom Employees' Provident Fund and Miscellaneous Provisions Act, 1952 is applicable.

Eligibility for membership:

The membership of EPS is mandatory for existing employees who were members of the ceased EFPS as also new employees who become members of EPF (including exempt provident funds) on or after 16.11.1995.

Contributions:

8.33% of employee's wages are contributed by the employer towards this Scheme out of the total contribution of employer to EPF of 10% or 12%, as the case may be. In addition 1.16% of employee's wages are paid by the Central Government. The wages for contribution purpose are restricted to a maximum of 15,000/ with effect from 1 September 2014 (6,500 earlier). Contribution in excess of the mandatory ceiling is not allowed under normal circumstances post this date.

Benefits:

The formula for Pensionable salary under the EPS95 is based on the average monthly PF salary in the 60 months before the EPS95 pension becomes payable instead of the earlier practice of one year. Also the Pensionable salary for the benefit calculation will now be calculated on a pro-rata service basis of the old PF salary ceiling of Rs6,500 and the new ceiling of Rs15,000 per month. This will only affect employees retiring in the next 60 months as everyone else will have 60 months' salary at the new ceiling level anyway.

A member shall be entitled to a:

- i. Superannuation Pension – if he has rendered the eligible service of 10 years or more and retires on attaining the age of 58 years,
- ii. Early Pension – if he has rendered eligible service of 10 years or more and retires or otherwise ceases to be in employment before attaining the age of 58 years.

Computation of Pension:

In case of a new entrant, the amount of monthly superannuation pension or early pension, as the case may be, shall be computed in accordance with the following formula:

Monthly member' pension = Pensionable Salary x Pensionable Service/ 70,

Where pensionable service shall mean the service for which contributions are received in respect of the employee in the Employees' Pension Fund and pensionable salary is the average of last 12 months' salary during the contribution period of service. Where the pensionable service is 20 years or more and the member superannuates on attaining the age of 58 years, a credit of 2 years is given in the Pensionable service.

In case of existing members, the superannuation or early pension shall be the aggregate of the monthly pension calculated as per above formula and the past service benefit depending upon the years of past service and the date of commencement of pension subject to a certain minimum amount.

Normally the pension is payable after attainment of 58 years irrespective of employer's policy of retirement age. However, a reduced pension after completing the age of 50 years may be paid in which case the amount of pension shall be reduced at the rate of 4% p.a. compounding for which the age falls short of 58 years.

Leaving before rendering eligible service of 10 years:

If a member does not render the eligible service of 10 years, he is entitled to a withdrawal benefit as per Table "D" or may opt for the Scheme certificate. Further existing members shall receive an additional amount for their contribution under Employees' Family Pension Scheme, 1971 computed as Table 'A' multiplied by the factors given in Table 'B'. The tables A, B, C & D are given at the end of the Scheme notified by the Government.

Permanent and Total Disablement:

A member on permanent and total disablement is entitled to an immediate pension without any reduction from the date following the date of disablement subject to a minimum pension of ` 250/- per month provided at least one month's contribution has been made to the Pension Fund.

Family Pension:

In case of death of a member while in service, the monthly widow/widower pension will be worked out as if the member had retired on the date of death subject to a minimum pension of `1000/-. Where death of the member takes place after commencement of pension to the member, the widow/widower pension will be equal to 50% of member's pension subject to a minimum pension of ` 1000/- per month. The widow/widower's pension is payable till her/his death or remarriage, whichever is earlier.

The monthly children pension for each child (maximum two children at a time) shall be equal to 25% of the amount of widow/widower's pension subject to a minimum of ` 250/-per month. It will be payable till 25 years of age of the child. The eligibility of children will be decided according to their seniority by age. Thus if there are more than two children, the pension will first be paid to two eldest of them. When pension for one of them is stopped (either due to death or on attaining 25 years age (or marriage in case of female child), the

next eldest child becomes eligible for family pension and so on. However, in case of disabled children, the pension admissibility remains for life and without restriction of linking to two children at a time.

[6]

(iv) Implication of revision of minimum pension under EPS:

The EPS is a defined benefit scheme and hence the cost would be uncertain and contributions would depend on payments and actual experience in future. The current contribution pattern is linked to the mandatory salary of INR 15,000 and hence is defined. Having a fixed contribution structure to fund such a scheme may not be sustainable in the medium to long term as there are many risks pertaining to investment, longevity, member data quality, administrative challenges etc...

While the sustainability of the enhanced pension under certain assumption scenarios would have been tested under the revised contribution basis, the impact of stress scenarios due to poor market conditions, additional membership due to the increase in mandatory salary ceiling or growth in number of covered establishments in a vibrant economic scenario, material increase in longevity due to medical advancements may put pressure on the affordability and sustainability of the increase.

For example, if the life expectancy increases by 3 years on average, then the cost of paying three extra years of a minimum monthly pension of INR 1,000, would put tremendous cash flow and cost pressure on the scheme.

[6]

[28 Marks]
